

Chiral Orientation of the Directed Heisenberg Frontier

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Abstract

The angular generation split of the projected Dirac residue E_Π is carried, on the directed Heisenberg frontier, by a chirally dressed cocycle signal $\Theta_\chi = \langle \sigma_L (2\pi/q) \Delta A_c \rangle_{\partial^+}$. The raw cocycle average and the maximal-locking bound $\theta_{\max}$ are computable from the arithmetic backend, but the signal itself depends on the chiral weight σ_L , whose origin lies in the Lorentzian spin-lift and not in the arithmetic frontier. We prove that σ_L cannot be any function of the real cascade generator $T(e)$: edge inversion and the residual reflection φ both act as $T \mapsto -T$, so every $f(T)$ has identical arrow-parity and φ -parity, whereas σ_L must be arrow-even and φ -odd. The resolution is that σ_L is the Weyl representation type, **2** versus **2̄**: group inversion preserves the representation (arrow-even) while complex conjugation exchanges it (φ -odd). In the inherited Heisenberg central-phase lift the chiral phase is the linear cyclotomic character $\omega_\chi(e) = \zeta_q^{\Delta A_c(e)}$, with no Gauss, Maslov, or $\sqrt{5}$ component; on the nonzero-cocycle support the natural branch satisfies $\sigma_L(e) = \text{sign } \Im \zeta_q^{\Delta A_c(e)} = \text{sign } \Delta A_c(e)$, giving $\rho_\chi = 1$ and $N_A = \theta_{\max} \neq 0$. This establishes [H-orient] in the inherited Heisenberg central-phase lift, not unconditionally: the only remaining obstruction is an additional spin-Galois phase outside the central character, in particular a $\sqrt{5}$ or ADE-spin factor orthogonal to ζ_q . The result is a conditional derivation of the frontier amplitude datum; it does not derive $u \neq 0$, which still requires the Schur transport of the polarisation onto the J_Π -odd generation sector.

Keywords: directed Heisenberg frontier; chiral orientation; Weil representation type; metaplectic central character; generation split; Schur complement; arithmetic orthogonality; epistemic stratification.

Contents

1	Introduction	3
2	Problem statement: $T(e)$ cannot define σ_L	4
3	The 2×2 obstruction	4
4	Representation-type resolution	5
5	The inherited central cyclotomic phase	6
6	Frontier overlap and the conditional signal	6
7	Residual caveat: the spin-Galois phase	7
8	Relation to the Schur transport	8
9	Status summary	8

1 Introduction

The projected Dirac residue admits the Schur-complement form $E_\Pi = -M^\dagger M$, and the inter-generation mass split is the J_3 -odd asymmetry of E_Π^2 on the gauge-singlet generation factor $\mathbb{C}_{\text{gen}}^3 \cong \text{Sym}^2(V)$ [7]. The oriented-frontier diagnostic of the recursive $Q11$ model reduces the angular part of this split to a single observable on the directed Cayley graph of $\text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ [6]. That diagnostic returns two computable quantities and one missing datum. The raw directed cocycle average vanishes,

$$\langle \Delta A_c \rangle_{\partial^+} = 0, \quad (1)$$

because the residual reflection φ pairs the outgoing frontier with itself and sends $\Delta A_c \mapsto -\Delta A_c$. The maximal-locking bound,

$$\theta_{\max} = \left\langle \frac{2\pi}{q} |\Delta A_c| \right\rangle_{\partial^+}, \quad (2)$$

is also computable and is stable under q . The signal that would turn the bound into an amplitude,

$$\Theta_\chi = \left\langle \sigma_L \frac{2\pi}{q} \Delta A_c \right\rangle_{\partial^+}, \quad |\Theta_\chi| \leq \theta_{\max}, \quad (3)$$

requires the chiral weight $\sigma_L(e) \in \{\pm 1\}$. The frontier backend does not supply σ_L : it is the surviving stub of the Lorentzian chiral lift, and the question [H-orient] is whether σ_L is non-orthogonal to $\text{sign } \Delta A_c$ on the outgoing frontier,

$$\rho_\chi = \frac{\langle \sigma_L \Delta A_c \rangle_{\partial^+}}{\langle |\Delta A_c| \rangle_{\partial^+}} \neq 0. \quad (4)$$

A numerical diagnostic in the inherited backend confirms the two computable quantities directly: with the campaign generators $\{\pm X, \pm Y\}$ of [1], the raw average is exactly zero, the symmetrised anti-bias control is exactly zero, and $\theta_{\max} \cdot q$ is q -invariant on the verified range $q \in \{61, 101, 151\}$. This validates the backend and the bound, but says nothing about σ_L . The present note derives σ_L from the inherited lift and thereby converts the bound into a signal under a precisely named phase hypothesis.

The discipline of the programme is respected throughout: no positive frontier run is read as a proof of $u \neq 0$. Every claim below carries an explicit status among *proved*, *structural*, *conditional*, and *open*.

2 Problem statement: $T(e)$ cannot define σ_L

The required parities are fixed by the oriented-frontier observable [6]. Under the residual reflection $\varphi = R_b \circ J_\Pi$ one has $\Delta A_c \circ \varphi = -\Delta A_c$ and $\gamma_5 \circ \varphi = -\gamma_5$, so the dressed signal $\gamma_5 \Delta A_c$ is φ -even and survives the re-pairing that kills the raw cocycle. Under cascade reversal the cocycle is arrow-odd, $\Delta A_c(e^{-1}) = -\Delta A_c(e)$, so the reversal control $\Theta_\chi^{(\partial^+)^{-1}} = -\Theta_\chi^{\partial^+}$ holds only if the chiral weight is transported rather than recomputed. The admissible σ_L must therefore satisfy

$$\sigma_L \circ \varphi = -\sigma_L \quad (\varphi\text{-odd}), \quad \sigma_L(e^{-1}) = \sigma_L(e) \quad (\text{arrow-even}). \quad (5)$$

By contrast $\text{sign } \Delta A_c$ is φ -odd and arrow-odd; the global identification $\sigma_L = \text{sign } \Delta A_c$ on all directed edges is therefore excluded, since it makes σ_L arrow-odd and breaks the reversal control.

3 The 2×2 obstruction

On the outer doublet $\{e_+, e_-\}$ of the J_Π -odd sector [3, Prop. 6.3], with the identifications

$$J_3 = \gamma_5 = \sigma_3, \quad J_\Pi = \sigma_1 K \text{ (antiunitary)}, \quad R_{\text{mix}} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = i\sigma_2, \quad (6)$$

the J_Π -odd projection of the cascade generator is

$$T(e) = \alpha(e) \sigma_3 + \mu(e) i\sigma_2 = \begin{pmatrix} \alpha & \mu \\ -\mu & -\alpha \end{pmatrix}, \quad \lambda_\pm = \pm \sqrt{\alpha^2 - \mu^2}, \quad (7)$$

where α is the real-split coefficient produced by the order $[X, Y] = Z$ and μ is the mixing coefficient [3, Prop. 6.5]. The conjugations $J_\Pi \sigma_3 J_\Pi^{-1} = -\sigma_3$ and $J_\Pi (i\sigma_2) J_\Pi^{-1} = -i\sigma_2$ give $J_\Pi T J_\Pi^{-1} = -T$, so T is J_Π -odd.

Lemma 1 (Parity collapse). *Let $U(e)$ be the lift of the oriented edge with $T(e) = \log U(e)$ restricted to the doublet. Edge inversion $U(e^{-1}) = U(e)^{-1}$ gives $T(e^{-1}) = -T(e)$, and the residual reflection gives $T(\varphi e) = \varphi T(e) \varphi^{-1} = -T(e)$. Hence both operations act as $(\alpha, \mu) \mapsto (-\alpha, -\mu)$, and every function $f(T)$ has identical arrow-parity and φ -parity.*

Proof. $U(e^{-1}) = U(e)^{-1} = \exp(-T(e))$ yields $T(e^{-1}) = -T(e)$. T is J_Π -odd and R_b is scalar on the chiral carrier [6], so $\varphi T \varphi^{-1} = -T$ and $T(\varphi e) = -T(e)$. A function $f(T)$ that is invariant (resp. anti-invariant) under $T \mapsto -T$ is therefore even (resp. odd) under both arrow inversion and φ simultaneously. $\square$

Theorem 1 (σ_L is not a function of $T(e)$). *There is no function f such that $\sigma_L = f(T(e))$ realises the admissible parities, because the admissible σ_L is arrow-even and φ -odd, while every $f(T)$ has equal arrow- and φ -parities.*

Status. **Proved.**

The spectral branch confirms the obstruction explicitly. The projector onto the positive eigenline $v_+ \propto (\mu, \lambda_+ - \alpha)$ gives

$$\text{Tr}(\gamma_5 P_{v_+}) = \frac{\mu^2 - (\lambda_+ - \alpha)^2}{\mu^2 + (\lambda_+ - \alpha)^2} = \frac{\lambda_+}{\alpha} = \text{sign}(\alpha) \sqrt{1 - \mu^2/\alpha^2}, \quad (8)$$

whose sign is $\text{sign}(\alpha)$, hence arrow-odd for every value of μ . Under the complexification $\text{mp}(2, \mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C})$ the real Cartan J_3 is the non-compact boost, so the spectral branch is exactly the boost sign, which is the excluded choice.

4 Representation-type resolution

The carrier V is the Weyl spinor of $\mathfrak{sl}_2(\mathbb{C})$, with the fundamental $\mathbf{2}$ (L) and its complex conjugate $\bar{\mathbf{2}}$ (R); the generation factor is $\text{Sym}^2(V)$, so $e_+ = L \otimes L$ and $e_- = R \otimes R$ [3, §6]. The lift acts on e_+ by $U \otimes U$ and on e_- by $\bar{U} \otimes \bar{U}$. The chiral weight is the representation type carried by the edge,

$$\sigma_L(e) = (\text{Weyl representation type of the lift of } e) \in \{+1 \text{ } (\mathbf{2}), -1 \text{ } (\bar{\mathbf{2}})\}. \quad (9)$$

Proposition 1 (Correct parities of the representation type). *Group inversion preserves the representation type, $\sigma_L(e^{-1}) = \sigma_L(e)$ (arrow-even), because U^{-1} lies in the same representation as U . Complex conjugation exchanges $\mathbf{2} \leftrightarrow \bar{\mathbf{2}}$, $\sigma_L(\varphi e) = -\sigma_L(e)$ (φ -odd), because φ contains the antiunitary J_{Π} .*

Status. **Structural.**

This is consistent with the real symplectic model. For a real lift $U \in \text{SL}(2, \mathbb{R})$ the restricted representation is self-conjugate: the distinction between $\mathbf{2}$ and $\bar{\mathbf{2}}$ is not a canonical label on the real carrier. Thus the effective chiral contrast carried by this real model is zero, which we denote schematically by $\sigma_L = 0$ before the complex Lorentzian lift is chosen. This matches the chiral symmetry of the real cascade, in which $\mu = 0$ [3, Rem. 6.4]. The representation type is the unique label distinguished by complex conjugation, which is precisely the operation separating φ from arrow inversion; this is why it can carry opposite parities under the two.

5 The inherited central cyclotomic phase

In the Schrödinger–Heisenberg convention of the exact Weil-block construction [1], the lift acts as

$$\rho_c(a, b, \gamma) f(x) = \zeta_q^{c(\gamma + b(x-a))} f(x - a), \quad \zeta_q = e^{2\pi i/q}, \quad (10)$$

with central character $\zeta_q^{c\gamma}$. The central increment of an oriented edge is the Heisenberg cocycle, so the edge phase is the linear cyclotomic character

$$\omega_\chi(e) = \zeta_q^{ca(g)s_b} = \zeta_q^{\Delta A_c(e)}. \quad (11)$$

The two chiral branches carry the conjugate central characters c and $q - c$, which is exactly the parity $c \leftrightarrow q - c$ realised by J_Π [3, §6].

Proposition 2 (Arithmetic nature of the inherited phase). *In this convention the edge phase is purely linear in the cocycle: the recovered exponent equals $ca(g)s_b$ with no quadratic residue, and $|\omega_\chi(e)| = 1$ on every edge. There is therefore no Gauss sum $\sqrt{q}$, no Maslov quadratic character, and no $\sqrt{5}$ amplitude in the inherited phase.*

Status. Proved within the inherited central cyclotomic phase convention. Verified directly on the campaign backend by exponent recovery and by termwise unit modulus.

The absence of a Fourier or Weyl generator in the Schrödinger–Heisenberg pipeline is the structural reason no metaplectic Gauss phase appears: the lift is a product of Heisenberg translations and modulations, whose only phase is the inherited central character.

6 Frontier overlap and the conditional signal

On the nonzero-cocycle support of the outgoing frontier, the natural branch of the inherited central phase is selected by the sign of its imaginary part,

$$\sigma_L(e) = \text{sign } \Im \zeta_q^{\Delta A_c(e)} = \text{sign } \Delta A_c(e), \quad e \in \partial^+, \quad (12)$$

since for a centred residue $r \in (-q/2, q/2)$ one has $\text{sign } \sin(2\pi r/q) = \text{sign } r$. This formula is an outgoing-frontier branch selection. For the paired inverted edge e^{-1} the chiral label is transported with the same branch,

$$\sigma_L(e^{-1}) := \sigma_L(e), \quad (13)$$

rather than recomputed from $\Delta A_c(e^{-1})$. This is the arrow-even transport required by the reversal control, realised as a label on the unoriented edge class with canonical radial representative, while complex conjugation still exchanges the two Weyl representation types and gives φ -oddness. The normalised overlap is then

$$\rho_\chi = \frac{\langle \sigma_L \Delta A_c \rangle_{\partial+}}{\langle |\Delta A_c| \rangle_{\partial+}} = +1, \quad (14)$$

and, restricted to the cocycle support, the locking is maximal.

Theorem 2 (Conditional orientation theorem). *In the inherited Heisenberg central-phase lift, the chiral weight σ_L equals $\text{sign } \Delta A_c$ on the nonzero-cocycle support of the outgoing frontier, transported arrow-even as a function of the unoriented edge. Consequently, in the frontier normalisation used in this diagnostic,*

$$\omega_\chi(e) = \zeta_q^{\Delta A_c(e)} \implies \rho_\chi = 1 \implies [\text{H-orient}] \implies N_A = \theta_{\max} \neq 0. \quad (15)$$

The equality with $\theta_{\max}$ is a maximal-locking statement for the inherited central-phase frontier channel. It is not an absolute calibration of the later spectral split.

Status. Proved within the inherited central cyclotomic phase convention. The overlap $\rho_\chi = 1$ is q -invariant on the verified range $q \in \{61, 101, 151\}$; the support fraction of the cocycle is itself q -invariant.

The reading is deliberately conservative: the campaign run validated the backend and the bound; the analysis of the inherited lift, and only that, converts the bound into a signal. No step infers $N_A \neq 0$ from a positive run.

7 Residual caveat: the spin-Galois phase

The result above is the phase inherited from the Heisenberg central character, which is the phase carried by the radial capacity pipeline [1]. The full Lorentzian spin-lift, with a genuine $\text{SL}(2, \mathbb{C})$ factor and the ADE-spin structure carrying $\sqrt{5}$, is not present in this pipeline. If the chiral sector carried an additional phase beyond the inherited central character, the dichotomy would reopen.

Remark 1 (The orthogonality alternative). Should the chiral branch acquire a phase in a Galois factor orthogonal to ζ_q , in particular a $\sqrt{5}$ or ADE-spin factor, then $\langle \sigma_L \text{sign } \Delta A_c \rangle_{\partial+}$ could vanish by character orthogonality [4],

yielding a clean no-go of the same type as the arithmetic obstruction governing the ADE case selection of the generation gate [3]. In that case the signal would exist operationally while the frontier amplitude vanished.

Status. **Open.** Whether the chiral spin-lift carries only the inherited central character, or an additional spin-Galois phase, is the single remaining obstruction; it is analytical and lies outside the present backend.

8 Relation to the Schur transport

The conditional orientation theorem fixes the frontier amplitude datum N_A , the normalisation entering the angular generation split [6, 5]. It does not, by itself, establish the polarisation transport that opens the mass split. The split lives in the projected Dirac residue through the Schur complement, and its J_Π -odd component on the generation sector is

$$\Pi_S D(\partial_s P|_0) D \Pi_S^* \Big|_{\text{gen}, J_\Pi\text{-odd}} \neq 0, \quad (16)$$

which is a distinct statement on the same chiral object [7]. The two stages are sequential: [H-orient] promotes the bound to a signal $N_A \neq 0$, and the Schur transport then promotes the signal to the spectral split $u \neq 0$ within the stratigraphic mass hierarchy [2].

Status. **Not derived here; requires Schur transport.**

9 Status summary

- $\sigma_L \neq f(T(e))$: **proved.**
- $\sigma_L = \text{Weyl representation type } (\mathbf{2} \text{ versus } \bar{\mathbf{2}})$: **structural.**
- $\omega_\chi(e) = \zeta_q^{\Delta A_c(e)}$ linear, no Gauss / Maslov / $\sqrt{5}$ component: **proved within the inherited central cyclotomic phase convention.**
- $\rho_\chi = 1$ on the cocycle support, hence $N_A = \theta_{\max} \neq 0$: **proved within the inherited central cyclotomic phase convention.**
- Absence of an additional spin-Galois phase: **open.**
- $u \neq 0$: **not derived here; requires Schur transport.**

The qualitative segment of [H-orient] is thus closed in the inherited lift, with the residual obstruction reduced to a single, sharply named analytical question: the spin-Galois phase of the full Lorentzian chiral lift.

References

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- [6] Jérôme Beau. The Oriented Frontier Observable: a Recursive Q11 Diagnostic for the Angular Generation Split, 2026. Fermionic matter sub-programme diagnostic note. The symmetric-capacity obstruction makes the raw shell observable of the angular generation split vanish identically ($\Theta_{\text{raw}} = 0$): the involution $g \mapsto g^{-1}$ pairs opposite central phases on symmetric BFS shells, erasing the oriented J_{Π} -odd component before

the radial recursion can act. This note defines the honest replacement: a frontier transfer observable on the directed outgoing edges $g \rightarrow gs$, whose central increment is the Heisenberg cocycle $a(g) s_b$, discrete counterpart of the BCH term $\frac{1}{2}[X, Y]$ and hence of the dynamical J_3 . The orientation is the recursive arrow $\partial_\tau = \log g$ from the projection origin. The first test is the existence of an oriented signal $\langle \Delta A_c \rangle_{\partial+S_m} \neq 0$, which may fail under residual frontier re-pairing; the discriminating anti-bias control is the symmetrised-frontier cancellation. No normalisation, no dictionary value, no ε enters; the note tests only whether the recursion produces a non-zero, bias-free oriented signal, doi:10.5281/zenodo.20601245.

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